

❖ Gaussian Process

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The **normal distribution** is also known as the **Gaussian distribution**, named after Carl Friedrich Gauss, who introduced it over 200 years ago in his work on modeling measurement errors in astronomy. The classical (univariate) normal distribution describes a single random variable. Its generalizations are:

- The **bivariate normal distribution**, which describes two random variables jointly.
- The **multivariate normal distribution**, which extends this to any finite number of random variables.
- A **Gaussian process**, which extends the idea even further, from finite-dimensional random vectors to infinite-dimensional objects, namely stochastic processes indexed by continuous time.

In other words:

Normal *distribution* $\Rightarrow$ Multivariate normal *distribution* $\Rightarrow$ Gaussian *process*.

Multivariate Normal Distribution

Let $X_1, X_2, \dots, X_k$ be random variables.

They are said to have a **multivariate normal distribution** if for every choice of real numbers $a_1, a_2, \dots, a_k$ the linear combination

$$a_1X_1 + a_2X_2 + \dots + a_kX_k$$

has a **univariate normal distribution**.

This definition emphasizes a key feature of normal distributions: *every linear combination is again normal*.

Mean Vector and Covariance Matrix

A multivariate normal distribution is completely determined by:

- The **mean vector**

$$\mu = (\mu_1, \dots, \mu_k) = (E(X_1), \dots, E(X_k)).$$

- The **covariance matrix**

$$V = (V_{ij})_{1 \leq i, j \leq k}, \quad V_{ij} = \text{Cov}(X_i, X_j).$$

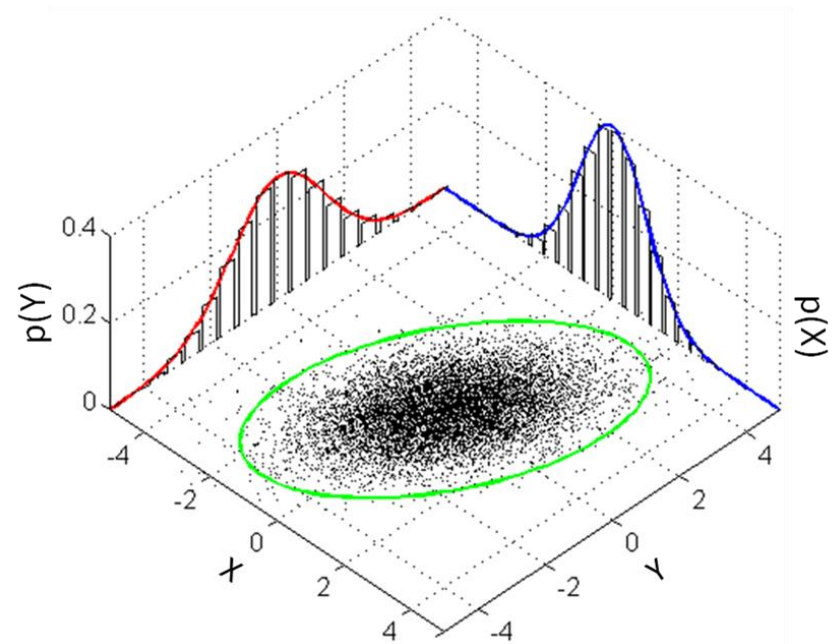
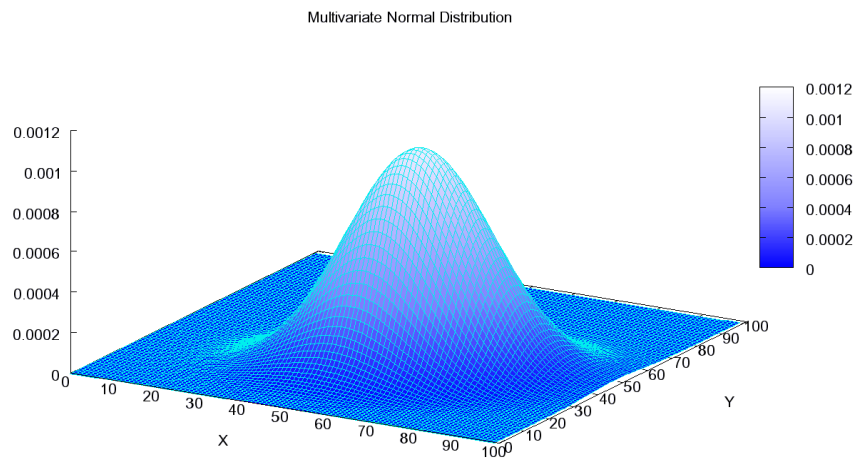
Thus,

$$V = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

Joint Density

If $X = (X_1, \dots, X_k)$ has a multivariate normal distribution with mean vector μ and covariance matrix V , then its joint density is

$$f(\vec{x}) = \frac{1}{(2\pi)^{k/2} |V|^{1/2}} \exp\left(-\frac{1}{2} (\vec{x} - \vec{\mu})^T V^{-1} (\vec{x} - \vec{\mu})\right),$$



Gaussian Process

A **Gaussian process** $(X_t)_{t \geq 0}$ is a continuous-time stochastic process such that for every finite collection of times

$$0 \leq t_1 < t_2 < \dots < t_n,$$

the random vector

$$(X_{t_1}, X_{t_2}, \dots, X_{t_n})$$

has a multivariate normal distribution.

Just as a multivariate normal distribution is determined by its mean vector and covariance matrix, a Gaussian process is completely determined by:

- The **mean function**

$$m(t) = E[X_t], t \geq 0,$$

- The **covariance function**

$$C(s, t) = \text{Cov}(X_s, X_t), \quad s, t \geq 0.$$

Gaussian Process and Brownian Motion

A stochastic process $(B_t)_{t \geq 0}$ is a **standard Brownian motion** if and only if it is a Gaussian process satisfying:

- **Initial condition**

$$B_0 = 0.$$

- **Mean function**

$$E(B_t) = 0 \text{ for all } t \geq 0.$$

- **Covariance function**

$$\text{Cov}(B_s, B_t) = \min\{s, t\} \text{ for all } s, t \geq 0.$$

- **Continuous sample paths**

$$t \mapsto B_t \text{ is continuous with probability } 1.$$

These four properties uniquely characterize standard Brownian motion among all Gaussian processes.

Proof Sketch: Brownian Motion is a Gaussian Process

Let $0 < t_1 < t_2 < \dots < t_k$. For constants $a_1, \dots, a_k$, consider

$$a_1 B_{t_1} + a_2 B_{t_2} + \dots + a_k B_{t_k}.$$

Using independent increments,

$$B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_k} - B_{t_{k-1}}$$

are independent normal random variables.

Rewrite:

$$\begin{aligned} a_1 B_{t_1} + \dots + a_k B_{t_k} &= (a_1 + \dots + a_k) B_{t_1} + (a_2 + \dots + a_k) (B_{t_2} - B_{t_1}) \\ &\quad + \dots + a_k (B_{t_k} - B_{t_{k-1}}) \end{aligned}$$

which is a linear combination of independent normal variables, hence itself normal. Therefore, Brownian motion is a Gaussian process.

Its mean and covariance are:

$$E(B_t) = 0, \quad \text{Cov}(B_s, B_t) = \min\{s, t\}$$

Converse: Gaussian Process $\Rightarrow$ Brownian Motion

Assume $(B_t)_{t \geq 0}$ is a Gaussian process satisfying:

$$B_0 = 0, \quad E(B_t) = 0, \quad \text{Cov}(B_s, B_t) = \min\{s, t\},$$

and with continuous paths. Then,

1. Stationary increments

For $s, t \geq 0$,

$$B_{t+s} - B_s \sim N(0, t)$$

since

$$\text{Var}(B_{t+s} - B_s) = (t + s) + s - 2s = t.$$

2. Independent increments

For $0 \leq q < r \leq s < t$,

$$\text{Cov}(B_r - B_q, B_t - B_s) = 0,$$

which implies independence because the increments are jointly normal.

